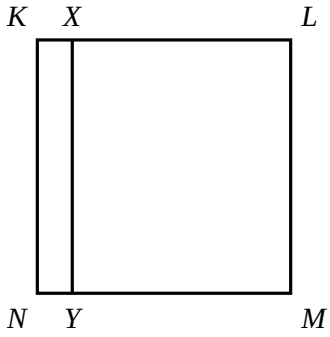
Math Kangaroo Practice

Grade 3-4 · 20 questions · harder · seed 39898

Part A: Each question is worth 3 points

1. *KLMN* is a square. Its side is 29 cm. The line *XY* cuts it into two rectangles, and *KX* = 4 cm.



How many centimetres is the perimeter of the square *KLMN* larger than that of the rectangle *XLMY*?

A. 4 cm ***B.*** 7 cm ***C.*** 8 cm ***D.*** 25 cm ***E.*** 58 cm

2. Chloe, Manon, Aisha, Sofia and Zoe each play a different instrument: the guitar, the trumpet, the xylophone, the violin and the drum. Sofia plays with sticks. Aisha plays with sticks. Zoe plays an instrument with strings. Aisha doesn't play the drum.
Who plays the violin?

A. Aisha ***B.*** Zoe ***C.*** Sofia
D. Manon ***E.*** Impossible to determine

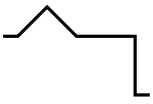
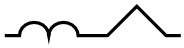
3. Hana, Noah, Chloe, Manon and Leo each play a different instrument: the drum, the xylophone, the flute, the trumpet and the harp. Hana does not blow into the instrument. Chloe plays with sticks. Leo plays with sticks. Manon doesn't play the trumpet. Chloe doesn't play the drum.
Who plays the xylophone?

A. Chloe ***B.*** Noah ***C.*** Hana
D. Leo ***E.*** Impossible to determine

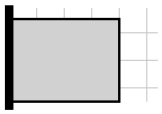
Part B: Each question is worth 4 points

4. Lena cut a strip of paper along this line.

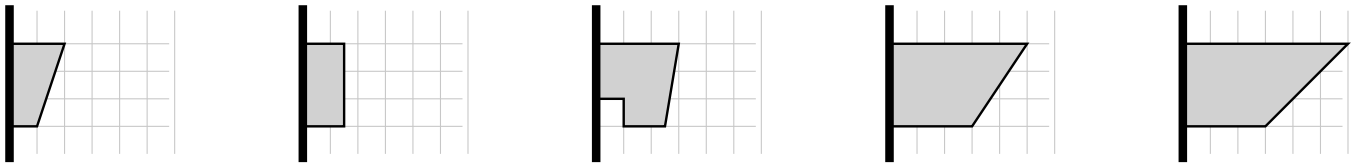
Which piece **can** come from the cut line?

A. ***B.*** ***C.***  ***D.*** ***E.*** 

5. From a window, Tom sees a rectangular flag on a pole. The wind turns and folds the flag, but the side sewn to the pole never moves. Tom draws the shadow of the flag on the wall at five different moments. Flat, the flag is 4 squares wide and 3 squares tall (see the picture). Some of the drawings are wrong.



the flag, flat

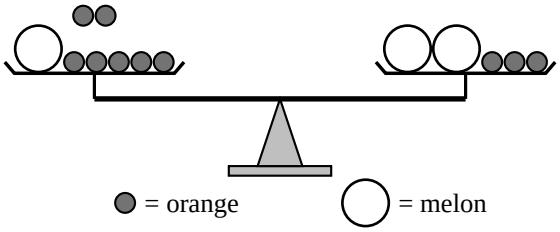


A B C D E

How many of the five drawings can be right?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

6. The balance is level: 1 melon and 7 oranges on one plate, 2 melons and 3 oranges on the other.



One melon weighs the same as how many oranges?

A. 5 oranges ***B.*** 4 oranges ***C.*** 3 oranges ***D.*** 2 oranges ***E.*** 1 orange

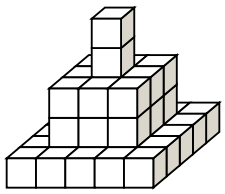
7. Tom has three cards with different digits. Two of them are 2 and 9; the third is face down. Using each card exactly once, with +, –, ×, ÷ and brackets, Tom gets 45.
Which of these digits can be on the hidden card?

A. 1 ***B.*** 5 ***C.*** 6 ***D.*** 7 ***E.*** 8

8. Omar has the cards 1, 2, 7 and 9. Omar must use every card exactly once, with +, –, ×, ÷ and brackets.
Which number can Omar *not* get?

A. 60 ***B.*** 54 ***C.*** 51 ***D.*** 49 ***E.*** 46

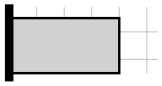
9. Leo built this pyramid out of identical cubes. Every layer is a full square, with no holes hidden inside.



How many cubes are in the pyramid?

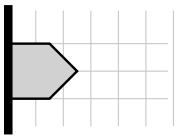
A. 13 ***B.*** 25 ***C.*** 35 ***D.*** 44 ***E.*** 45

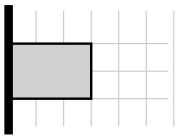
10. From a window, Mia sees a rectangular flag on a pole. The wind turns and folds the flag, but the side sewn to the pole never moves. Mia draws the shadow of the flag on the wall at five different moments. When there is no wind at all, the flag hangs flat and is 4 squares wide and 2 squares tall (see the picture).

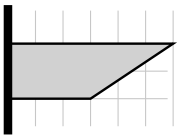


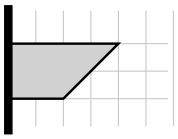
the flag, flat

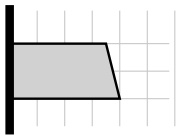
Which shadow cannot be right?

A. 

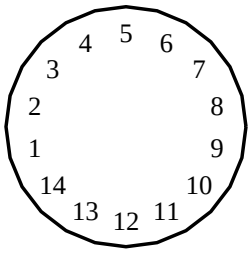
B. 

C. 

D. 

E. 

11. A round plate with the numbers 1 to 14 around its edge is cracked into 2 pieces. The sums of the numbers in the pieces are two consecutive numbers.

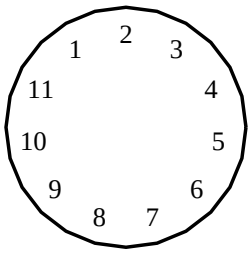


What is the largest of these sums?

A. 53 ***B.*** 52 ***C.*** 51 ***D.*** 42 ***E.*** 35

Part C: Each question is worth 5 points

12. A round plate with the numbers 1 to 11 around its edge is broken by one straight crack into two pieces. The sums of the numbers on the two pieces are equal.



Where does the crack go?

A. between 2 and 3, and between 8 and 9

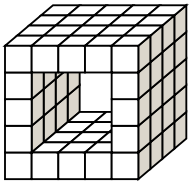
B. between 3 and 4, and between 6 and 7

C. between 1 and 2, and between 9 and 10

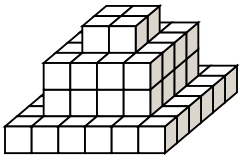
D. between 8 and 9, and between 11 and 1

E. between 1 and 2, and between 6 and 7

13. Robert has the tunnel on the left. Robert takes it apart to build the pyramid on the right, but there are not enough cubes. Nothing is hollow inside the pyramid.



tunnel

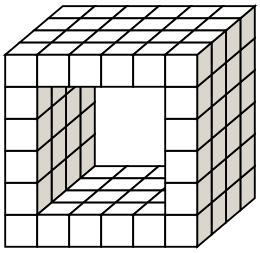


pyramid

How many more cubes does Robert need?

A. 56 ***B.*** 24 ***C.*** 10 ***D.*** 9 ***E.*** 8

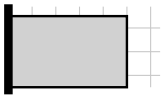
14. Robert's tunnel is 4 cubes long. The hole goes straight through. Robert makes the tunnel longer, keeping the same front, until it is 6 cubes long.



How many cubes does Robert have to add?

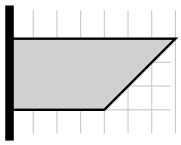
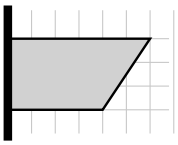
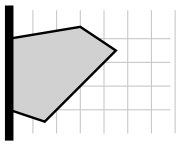
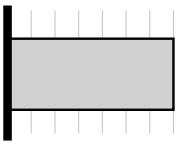
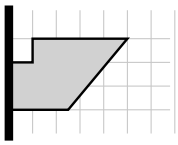
A. 120 ***B.*** 72 ***C.*** 60 ***D.*** 49 ***E.*** 40

15. From a window, Nina sees a rectangular flag on a pole. The wind turns and folds the flag, but the side sewn to the pole never moves. Nina draws the shadow of the flag on the wall at five different moments. Flat, the flag is 5 squares wide and 3 squares tall (see the picture). Four of the drawings are wrong.

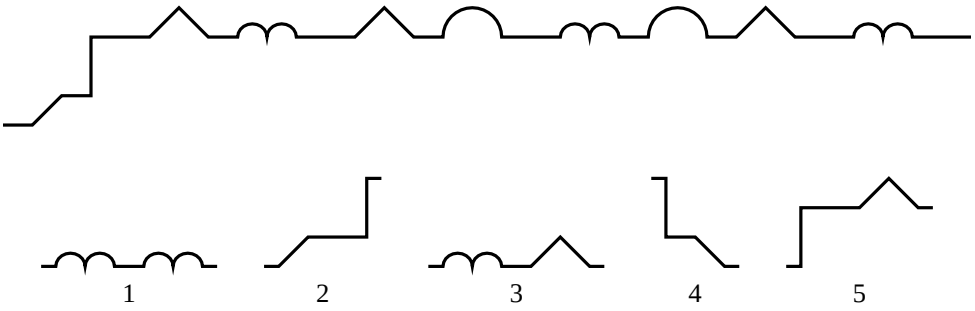


the flag, flat

Which drawing can be right?

A. 
B. 
C. 
D. 
E. 

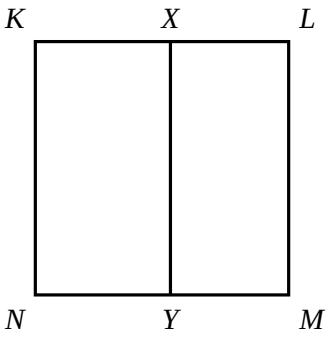
16. Lena cut a strip of paper along this line.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 5

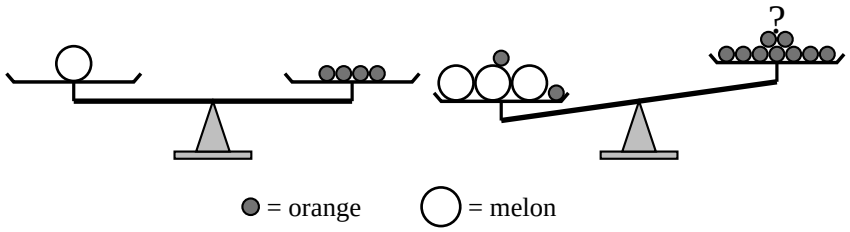
17. The square *KLMN* is cut into two rectangles. The perimeter of *KXYN* is 46 cm and the perimeter of *XLMY* is 44 cm.



What is the perimeter of the square *KLMN*?

A. 90 cm ***B.*** 75 cm ***C.*** 67 cm ***D.*** 65 cm ***E.*** 60 cm

18. The first balance is level. On the second balance there are 3 melons and 2 oranges on the left and 9 oranges on the right.



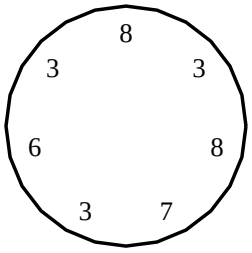
How many oranges must be added to the right plate to make it level?

A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 14

19. Manon, Nicolas, Tom, Sofia and Fabienne were born in different months: June, January, April, October and August. Fabienne has a birthday in summer. Nicolas was born in spring. Sofia was born in a month that starts with J. Tom was born in winter.
Which statement is **not** true?

A. Tom was born in January ***B.*** Fabienne was born in August ***C.*** Nicolas was born in April
D. Fabienne was born in April ***E.*** Sofia was born in June

20. Numbers are written around a round plate as shown. Tom wants to break the plate with one straight line so that the two pieces have the same sum of numbers.



In how many different ways can he do it?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

Answer Key

1.	**C**	**2.**	**E**	**3.**	**A**	**4.**	**E**	**5.**	**C**	**6.**	**B**
7.	**D**	**8.**	**C**	**9.**	**E**	**10.**	**C**	**11.**	**A**	**12.**	**A**
13.	**E**	**14.**	**E**	**15.**	**C**	**16.**	**B**	**17.**	**E**	**18.**	**D**
19.	**D**	**20.**	**B**								

Solutions

1. C (8 cm) — *XLMY* has sides 25 and 29 cm, perimeter 108 cm. Square: 116 cm. Difference 8 cm (= 2 × 4). [2002 Q6 · square_minus_strip]

2. E (Impossible to determine) — The clues allow more than one answer: Chloe plays the violin or Manon plays the violin or Zoe plays the violin. So it is impossible to determine. [2002 Q18 · who_has]

3. A (Chloe) — The only possible matching: Hana – harp; Noah – trumpet; Chloe – xylophone; Manon – flute; Leo – drum. [2002 Q18 · who_has]

4. E — Only this piece is found in the line. The others don't fit — A: its step is taller than in the line; B: it is a mirror image of a part of the line; C: it steps down where the line steps up; D: it is a mirror image of a part of the line. [2002 Q7 · can_belong]

5. C (2) — A drawing is wrong if part of the pole edge is bare or if it reaches more than 4 squares from the pole. Wrong: C, D, E. So 2 can be right. [2002 Q15 · count_possible]

6. B (4 oranges) — Take 1 melon and 3 oranges off both plates. Left: 1 melon = 4 oranges. [2002 Q9 · both_sides]

7. D (7) — (7 – 2) × 9 = 45. The other digits do not work with 2 and 9. [2002 Q21 · hidden_card]

8. C (51) — 1 + (7 – 2) × 9 = 46; 1 × (9 – 2) × 7 = 49; (1 + 7 – 2) × 9 = 54; 7 × 9 – 2 – 1 = 60. But with 1, 2, 7 and 9 nothing gives 51 (the closest are 50 = 1 + (9 – 2) × 7 and 52 = (7 – 1) × 9 – 2). [2002 Q21 · which_cannot]

9. E (45) — Count layer by layer, from the bottom: 5 × 5 + 3 × 3 × 2 + 1 × 1 × 2 = 45. [2002 Q20 · count_pyramid]

10. C — Turning or folding can only make the flag look narrower. Picture C reaches more than 4 squares away from the pole, which is wider than the flag. [2002 Q15 · too_wide]

11. A (53) — All the numbers add up to 105. Two consecutive numbers that add up to 105: 52 + 53 = 105. The largest is 53. [2002 Q11 · largest_sum]

12. A (between 2 and 3, and between 8 and 9) — Total 66, so each piece has 33: 3+4+5+6+7+8 = 33 and 9+10+11+1+2 = 33. [2002 Q11 · straight_crack]

13. E (8) — Tunnel: 16 × 4 = 64 cubes. Pyramid: 6 × 6 + 4 × 4 × 2 + 2 × 2 = 72. Missing: 72 – 64 = 8. [2002 Q20 · not_enough]

14. E (40) — Each slice of the tunnel has 6 × 6 – 4 × 4 = 20 cubes. 2 new slices: 20 × 2 = 40. [2002 Q20 · longer_tunnel]

15. C — Check two things: the pole edge must be covered all the way, and nothing may reach more than 5 squares from the pole. Only picture C passes both checks. [2002 Q15 · only_one_right]

16. B (1) — Only piece 5 fits. Not: 1 (it has two small bumps where the line has a round dome); 2 (its flat part between the two features has the wrong length); 3 (its flat part between the two features has the wrong length); 4 (it is a mirror image of a part of the line). [2002 Q7 · how_many_fit]

17. E (60 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 46 + 44 = 90 cm = 6 sides. Side = 90 ÷ 6 = 15 cm, perimeter 4 × 15 = 60 cm. [2002 Q6 · two_perimeters]

18. D (5) — One melon = 4 oranges, so the left plate is worth 3 × 4 + 2 = 14 oranges. The right plate has 9, so add 14 – 9 = 5. [2002 Q9 · what_to_add]

19. D (Fabienne was born in April) — The only possible matching: Manon – October; Nicolas – April; Tom – January; Sofia – June; Fabienne – August. [2002 Q18 · which_not_true]

20. B (1) — Each half must have 19. The ways: 8+3+8 | 7+3+6+3. [2002 Q11 · count_ways]